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Chapter 1

Briefing Document

What is Business Statistics?

- **Definition:** Business statistics is defined as a “collection of procedures and techniques used to convert data into meaningful information in a business environment”.
- **Key Processes:** It involves two main types of statistical procedures:
- **Descriptive Statistics:** These techniques are used to summarise and describe data (e.g. using charts, graphs, tables and numerical measures). This includes measures such as calculating an average. The document gives the following mathematical example: “Sum of all data values / Number of data values = Average = Sum of X_i / N ”
- **Inferential Statistics:** These tools and techniques are used to draw inferences and conclusions from data. Examples given include estimation and hypothesis testing. The document explains this with the following examples: “Estimate the average family income of all families in a city based on the average income of a sample of families in that city.” and “Use sample evidence to test the claim that the average family income exceeds \$45,000 per year.”

Data Collection Methods

Written Questionnaires and Surveys:

- These are generally less expensive than telephone surveys.
- They can more easily incorporate open-ended questions.
- They require careful formatting and a motivational statement of purpose.

Open-ended questions: Allow for free responses using a "any value, words, or statements of their own choosing". An example is given: "Under what circumstances would you favor a reduction in the Federal long-term capital gains tax rate?".

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- **Advantages:** Standardised questions, electronic data capture (if online), potentially more questions.
- **Disadvantages:** Potential for low response rates (5-10%), delayed responses, no clarification possible.
- **Typical Applications:** Employee surveys, course/professor evaluations, customer satisfaction studies, academic research.

Telephone Surveys:

- Typical applications include political polls, market research, customer satisfaction and healthcare studies.
- **Closed-end questions:** Require the respondent to choose from a predefined list of choices. An example given is: “What is your preferred rental car company?”.
- **Demographic questions:** Gather information about the respondents’ attributes and backgrounds (e.g. age and gender).
- **Issues:** Screening calls, respondent availability, cell phone dialing, short survey length, and trained callers needed.

Experiments:

- An experiment is defined as a process with an uncertain outcome.
- Experimental design is a plan to define the variable of interest.

Direct Observation and Personal Interviews:

- **Direct observation:** Data is recorded as events take place. It can be subjective and time-consuming. It can be used to determine customer product preference, assess compliance with rules or determine seatbelt usage. Challenges include avoiding the “Hawthorne effect”, maintaining consistency between observers, observer bias, and cost.
- **Personal interviews:** Can be structured (scripted questions) or unstructured (broad questions guiding the discussion). These can be used to measure customer satisfaction, assess supplier qualification and measure health status.

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- **Structured Interviews** have a specific set of predetermined questions, asked in the same order to all respondents, with no follow-up of answers.
- **Unstructured Interviews** do not have a predetermined set of questions. They begin with a general question and allow the discussion to go in any direction. The interviewer probes respondents for more information.
- **Challenges:** Interviewer bias, accurate response capture, time requirements, limited number of interviewees, training and consistency.
- **Other Data Collection Techniques:**
- **Focus Groups:** Group interviews with a facilitator to gather perspectives and opinions. Responses are then categorised for analysis. Participants often have common attributes.
- **Documents and Records:** Using existing (secondary) data from sources like databases, financial records and annual reports. This is potentially less expensive than generating new (primary) data but may be incomplete.
- **Bar Codes:** Provide product information, automatically captured by a scanner.
- **RFID (Radio Frequency Identification Device):** Transmitting devices attached to products; receivers read the data without needing a line of sight. Holds more data than a barcode and performs better in harsh environments but the signals can be compromised.

Data Collection Issues

- Interviewer Bias
- Nonresponsive Bias
- Selection Bias
- Observer Bias
- Measurement Error
- Internal Validity
- External Validity
- Data Accuracy

Populations, Samples, and Sampling Techniques

- **Population:** The entire set of objects or individuals of interest.
- **Sample:** A subset of the population.

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- **Census:** An enumeration of all measurements taken from the entire population.
- **Parameters vs. Statistics:**
- **Parameters:** Descriptive numerical measures calculated from a *population* (e.g. the average yards gained per play by all NFL teams in the 2016 season).
- **Statistics:** Descriptive numerical measures calculated from a *sample* (e.g. the average credits taken by a sample of students at a university).
- **Sampling Techniques:** Divided into two broad categories:
- **Statistical Sampling (Probability Sampling):** Uses chance selection techniques to select items. This allows every item in the population to have a known or calculable chance of being included in the sample.
- **Nonstatistical Sampling:** Uses methods based on convenience, judgement or other non-chance processes.
- **Nonstatistical Sampling Techniques:**
- **Convenience Sampling:** Collecting data in the most convenient way (e.g. interviewing students entering the library).
- **Judgement Sampling:** Selecting individuals based on who is likely to provide useful information (e.g. interviewing largest suppliers).
- **Ratio Sampling:** Selecting a sample with the same proportions as key segments in the population (e.g. ensuring a sample has the same percentages of women and college graduates).
- **Statistical Sampling Techniques:**
- **Simple Random Sampling:** Every possible sample of a given size has an equal chance of being selected. Can be with or without replacement. Can be generated using random number tables or computer software like Excel.
- **Stratified Random Sampling:** Dividing the population into subgroups (strata) based on a common characteristic and then selecting a simple random sample from each subgroup and combining. Advantageous when data is similar within the strata, reducing the size of the sample needed.
- **Systematic Random Sampling:** Dividing the population into groups and selecting one individual from the first group, then every k th individual.
- **Cluster Sampling:** Dividing the population into "clusters" (each representative of the whole population), then randomly selecting some clusters for inclusion in the sample.

Data Types and Data Measurement Levels

- **Quantitative Data:** Inherently numerical (e.g. height, age).
- **Discrete:** Countable (e.g. number of children).
- **Continuous:** Can take any value within a range (e.g. weight, volume).
- **Qualitative Data:** Categorical (e.g. marital status, eye colour).
- **Time-Series Data:** Observed at successive points in time (e.g. daily stock prices over a year).
- **Cross-Sectional Data:** Observed at a fixed point in time (e.g. bank data about loan customers).
- **Data Measurement Levels:**
- **Nominal Data:** Categorical data without order or ranking (e.g. Student ID, favourite colour, gender, marital status).
- **Ordinal Data:** Categorical data with a defined order or ranking (e.g. College Class Standing, Age Group, Satisfaction Level).
- **Interval Data:** Numerical data where the intervals between values are meaningful, but there is no true zero point (e.g. Temperature).
- **Ratio Data:** Numerical data with a meaningful zero point and meaningful ratios (e.g. weight, time, pay rate, interest rates).

Categorising Data

- Identify each factor in the data set.
- Determine if the data is time-series or cross-sectional.
- Determine which factors are quantitative or qualitative.
- Determine the level of measurement for each factor.

FQA

1. What is the core purpose of business statistics?

Business statistics is essentially a set of procedures and techniques designed to transform raw data into meaningful information within a business context. This involves describing data, drawing inferences, making estimations, and testing hypotheses, all to aid decision-making.

2. What are the two broad categories of statistical procedures?

The two main branches are *descriptive statistics* and *inferential statistics*. Descriptive statistics deals with summarising and describing data through charts, graphs, tables, and numerical measures like averages. Inferential statistics involves making predictions and inferences about a larger population based on a sample of that population.

3. What are some key methods for collecting data?

Several techniques are available for data collection, including written questionnaires and surveys, telephone surveys, direct observation, and personal interviews. Experiments are also a method, as are focus groups, document reviews, and the use of bar codes or RFID for product information.

4. How do telephone surveys differ from written questionnaires?

Telephone surveys typically use structured, closed-ended questions and can obtain immediate responses, though they often need to be brief due to respondent fatigue. Written questionnaires, however, can include more complex questions, allow for open-ended responses, are generally less expensive, but suffer from potentially low response rates and lack the chance to clarify questions in real-time.

5. What are the key differences between structured and unstructured interviews?

Structured interviews use a predetermined set of questions, administered in the same order to all respondents with no follow-up questions or elaboration. Unstructured interviews, on the other hand, begin with general questions and allow the discussion to evolve based on the respondent's answers, which facilitates more detailed exploration.

6. What are the differences between a population, a sample, and a census?

A *population* is the entire group of individuals or objects of interest. A *sample* is a subset of that population. A *census* is an enumeration of the entire population, collecting data from every member.

7. What are some of the various sampling techniques, and what distinguishes statistical from non-statistical approaches?

Sampling techniques include non-statistical methods such as convenience, judgement, and ratio sampling, which do not use chance and can have inherent biases, and statistical methods,

which use chance to give a known probability to each element of the population to be included in the sample. These statistical methods include simple random, systematic, stratified and cluster sampling.

8. What are the different levels of data measurement, and can you give examples?

Data can be classified into different measurement levels: *nominal* data, which are categorical with no inherent order (e.g., gender, marital status); *ordinal* data, which have a defined order but the intervals between them might not be equal (e.g., class standing, age group ranges); and *interval/ratio* data which are numerical and allow for mathematical operations, where an interval scale doesn't have a true zero value (e.g. temperature) but a ratio scale does (e.g. weight, time, pay rate, interest rates). Understanding these levels is crucial for proper statistical analysis.

Quiz

Instructions: Answer the following questions in 2-3 sentences each.

1. What is the fundamental purpose of business statistics?
2. Briefly explain the difference between descriptive and inferential statistics.
3. List three common methods for collecting data.
4. What are the key challenges associated with telephone surveys?
5. What is the main advantage of using written questionnaires compared to telephone surveys?
6. Explain the "Hawthorne effect" in the context of direct observation.
7. What is the difference between a population and a sample?

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8. Describe the characteristics of non-statistical sampling methods?
9. Why is stratified random sampling often more efficient than simple random sampling?
10. What distinguishes quantitative data from qualitative data?

Essay Questions

Instructions: Answer each of the following questions in essay format.

1. Discuss the various data collection methods outlined in the source material, comparing their advantages, disadvantages, and typical applications within business settings.
2. Explain the difference between statistical and non-statistical sampling methods. In your explanation, provide examples of where you would utilize each method.
3. Critically assess the potential sources of bias that can arise during the data collection process, and suggest strategies that might be used to mitigate these problems.

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4. Describe the differences between time-series and cross-sectional data. Explain the importance of understanding these different formats when analysing data.

5. Elaborate on the four levels of data measurement, providing real-world examples of each and explaining how the level of measurement impacts the statistical analysis that can be performed.

Quiz Answer Key

1. Business statistics is the collection of procedures and techniques used to convert data into meaningful information in a business environment. It allows businesses to understand trends, make informed decisions, and improve overall performance.
2. Descriptive statistics focuses on summarising and describing data using tools like charts, graphs and numerical measures. Inferential statistics, on the other hand, uses sample data to make inferences and predictions about a larger population.

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3. Three common methods for data collection include: written questionnaires/surveys, experiments, and telephone surveys. Direct observation and personal interviews are also common.
4. Key challenges associated with telephone surveys include low response rates, the potential for respondents to screen calls or hang up, and the necessity for trained callers. Additionally, cell phone numbers must be manually dialed.
5. Compared to telephone surveys, written questionnaires are often less expensive to administer and can more easily accommodate open-ended questions, and online questionnaires can directly capture data electronically.
6. The "Hawthorne effect" refers to changes in behaviour of those being observed, simply because they know that they are being observed. This can lead to biased data in direct observation studies.
7. A population is the complete set of all objects or individuals of interest, whereas a sample is a subset of that population that is selected for study. Samples are used to represent the larger population and draw inferences from them.
8. Non-statistical sampling methods use convenience, judgment, or other non-chance processes to select samples. This can lead to biased results as the selection is not random.
9. Stratified random sampling divides the population into subgroups with common characteristics before sampling, ensuring representation of each group in the sample. By reducing variability within each subgroup, this often provides a more precise estimate than simple random sampling and can reduce overall sample size.
10. Quantitative data consists of numerical measurements that can be either discrete or continuous. Qualitative data, in contrast, refers to categorical data where measurement is based on category rather than numbers, such as marital status or eye colour.

Glossary of Key Terms

1. **Business Statistics:** A collection of procedures and techniques used to convert data into meaningful information within a business environment.
2. **Descriptive Statistics:** Procedures and techniques designed to describe data in a summarised format, such as charts, graphs, and numerical measures.
3. **Inferential Statistics:** Tools and techniques that help decision-makers to draw inferences and make predictions from a set of data.
4. **Experiment:** A process that produces a single outcome whose result cannot be predicted with certainty, often involving controlled manipulations of variables.
5. **Experimental Design:** A plan for performing an experiment, carefully defining the variables of interest and the method of control and observation.
6. **Telephone Survey:** A data collection method that involves asking respondents questions via phone call, with potential challenges in low response rates and screening.
7. **Written Questionnaire:** A data collection tool with questions in written format, often requiring careful formatting and a statement of purpose.
8. **Open-End Questions:** Questions that allow respondents to freely respond with their own words, statements, or values, providing richer, but sometimes harder-to-analyse, data.
9. **Closed-End Questions:** Questions that require respondents to select from a defined list of options, offering easily quantifiable responses.
10. **Demographic Questions:** Questions relating to the respondents' characteristics, backgrounds, and attributes, providing insights into who is providing the data.
11. **Direct Observation:** A data collection method where researchers physically observe and record what takes place in a process, offering first-hand information.
12. **Personal Interview:** A data collection method involving a structured or unstructured conversation to obtain detailed information from the respondent.
13. **Structured Interview:** An interview with a predetermined set of questions asked in the same order to all respondents, limiting flexibility and follow up.
14. **Unstructured Interview:** An interview that begins with a general question, allowing the conversation to flow based on the respondent's answers, providing a more in-depth response.

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15. **Focus Group:** A group interview facilitated to gain data about the group's perspectives and opinions, often used in qualitative research.
16. **Documents and Records:** The study of existing data such as databases and reports, providing access to potentially less expensive and readily available information.
17. **Interviewer Bias:** Bias introduced by an interviewer's actions, body language, opinions or wording of questions, affecting the quality of the data collected.
18. **Nonresponsive Bias:** Bias introduced when respondents refuse to participate or do not respond to surveys, leading to inaccurate results.
19. **Selection Bias:** Bias introduced when the method of selecting participants results in a sample that is not representative of the population.
20. **Observer Bias:** Bias introduced by the observer's preconceived notions or behaviour, affecting the interpretation of observed data.
21. **Measurement Error:** Discrepancies between the true value and the measured value, which can occur during the data gathering process.
22. **Internal Validity:** The degree to which a research design accurately measures the intended variables without the influence of confounding factors.
23. **External Validity:** The degree to which the findings of a study can be generalised to other populations, settings, and conditions.
24. **Population:** The complete set of all individuals or objects of interest in a study, encompassing the total target of a study.
25. **Sample:** A subset of the population selected for observation or analysis, representing a portion of the larger group of interest.
26. **Census:** An enumeration of the entire set of measurements taken from the entire population, unlike a sample which is only a subset.
27. **Parameter:** A descriptive numerical measure computed from an entire population, representing the true value being measured in a population.
28. **Statistic:** A descriptive numerical measure computed from a sample, representing an estimation of the parameter in the population.
29. **Statistical Sampling:** Sampling methods that use random selection, giving every item in the population a known or calculable probability of being included.
30. **Nonstatistical Sampling:** Sampling methods that are not based on random selection and may include convenience, judgement, or other non-random selection processes.

31. **Convenience Sampling:** A nonstatistical method of collecting data in the most convenient manner for the researcher, with no regard for randomness or representation.
32. **Judgement Sampling:** A nonstatistical method of sample selection based on the researcher's judgement of who in the population is most likely to provide the desired information.
33. **Ratio Sampling:** A sample where a sample is selected to proportionally represent segments within the population, ensuring that subgroup sizes are proportionate.
34. **Simple Random Sampling:** A statistical method of sampling where every possible sample of a given size has an equal chance of being selected.
35. **Stratified Random Sampling:** A statistical method that divides the population into subgroups (strata) based on some common characteristic and takes a simple random sample from each stratum.
36. **Systematic Random Sampling:** A statistical method where the sample is obtained by selecting every kth individual from an ordered list of the population.
37. **Cluster Random Sampling:** A statistical method that divides the population into several clusters and then randomly selects some of these clusters for inclusion in the study, sampling from within these clusters.
38. **Quantitative Data:** Data that are numerical and represent measurements or counts, providing values in measurable terms.
39. **Qualitative Data:** Data that is categorical and represents attributes or qualities, describing things in categories rather than numbers.
40. **Time-Series Data:** A set of consecutive data values observed at successive points in time, tracking changes over a period.
41. **Cross-Sectional Data:** Data values observed at a fixed point in time, presenting a snapshot of a specific moment.
42. **Nominal Data:** A type of categorical data where the values are names or labels with no inherent order or ranking (e.g. gender, marital status).
43. **Ordinal Data:** A type of categorical data where the values have a meaningful order or ranking, but not a consistent interval between values (e.g. student satisfaction).
44. **Interval Data:** A type of numerical data where the values have a meaningful order with consistent intervals between values, but there is no true zero point (e.g. temperature).

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45. **Ratio Data:** A type of numerical data where the values have a meaningful order, consistent intervals, and a true zero point, allowing ratios to be meaningfully calculated (e.g. weight, income).

Chapter 2

Briefing Document

Key Themes:

1. **Frequency Distributions:** The chapter introduces frequency distributions as a fundamental way to summarise data by showing the number of observations within distinct categories or classes. This applies to both *discrete* (countable) and *continuous* (uncountable) data.
 - **Discrete Data:** Examples include student ages, product categories purchased, number of late deliveries, and touchdowns scored per game. Frequency distributions are created by listing possible values and then counting their occurrences. "During 13.3% of the months, the company had 0 late deliveries".
 - **Continuous Data/Grouped Data:** Continuous data (e.g., weight, length, time) or discrete data with many outcomes are often grouped into classes. Classes must be mutually exclusive, all-inclusive, and ideally of equal width. The number of classes can be determined using the " $2^k \geq n$ rule", where 'k' is the number of classes and 'n' is the number of data values, with a recommendation of between 5 and 20 classes, a minimum class width can then be calculated.
 - **Relative Frequency:** This expresses the proportion of total observations in a given category or class. Example: "9% of the people in the sample report that they read the newspaper 2 days per week."
 - **Cumulative Frequency/Cumulative Relative Frequency:** These display the number or proportion of observations less than or equal to the upper limit of each class.
2. **Histograms:** These are graphs that display a frequency distribution, with the horizontal axis representing possible outcomes and the vertical axis showing their frequencies. Key features of histograms include:
 - Displaying the centre, spread, and shape of the data. The chapter visually illustrates how histograms can differ in these aspects even if some factors are kept constant, by showing

"Same Spread – Same Shape – Different Centers" and "Same Center – Different Spread – Different Shape".

- No gaps between bars (unlike bar charts).

3. **Joint Frequency Distributions:** These are used when data are characterised by multiple variables, presenting frequencies for combinations of categories or classes across these variables. Example: how many customers paid by charge and used a specific garage, " 40 Customers Paid By Charge and Used Garage 1 ".

4. Categorical Data Visualisation:

- **Bar Charts:** These display categorical data with bars whose lengths/heights correspond to frequencies or percentages. Bars can be vertical or horizontal.
- **Pie Charts:** Circular charts where slices represent categories, with the size of each slice proportional to the category's contribution to the total. Pie charts should not be used when the categories don't relate to the proportion of a whole.

5. Quantitative Data Visualisation:

- **Stem and Leaf Diagrams:** A method for presenting quantitative data in a way that allows both a visual representation of the shape of the data and also keeps the original values.
- **Line Charts:** Display time-series data, with time on the horizontal axis and the variable of interest on the vertical axis, and lines connecting points. Line charts can be used to compare multiple variables, sometimes with improved versions using two vertical axes.
- **Scatter Diagrams (Scatter Plots):** These graphs show the relationship between two quantitative variables, with one variable on each axis. They help to visualize the relationship of a Dependent and Independent variable.

6. **Pareto Charts:** These are specialised bar charts that prioritise categories from highest to lowest, used to identify the "vital few" contributing factors from the "trivial many". "Used to Separate the “Vital Few” from the “Trivial Many”".

Key Ideas and Facts:

- **Data Types:** The importance of distinguishing between discrete (countable) and continuous (uncountable) data is emphasized, as different methods are often appropriate for each.
- **Class Intervals:** When grouping continuous data, the class intervals should be mutually exclusive, all-inclusive, and ideally of equal width to avoid distortion. The optimal number of classes should be considered carefully: "Too Many (Narrow class intervals) – May yield a very jagged distribution with gaps from empty classes" and "Too Few (Wide class intervals) – May compress variation too much and yield a blocky distribution".
- **Visual Interpretation:** The chapter emphasizes not only the construction of charts and diagrams but also their interpretation, highlighting that they provide insights into the data's center, spread, shape, and relationships.
- **Appropriate Graph Usage:** Certain graph types are better suited for different kinds of data. Pie charts, for example, should only be used when categories relate to a whole, as illustrated by the "Incorrect Use of a Pie Chart" example.

Outcomes (as Stated in the Document):

The learning outcomes listed at the start of the chapter are clearly covered throughout the excerpts:

- Constructing frequency distributions manually and with computers.
- Constructing and interpreting frequency histograms.
- Developing and interpreting joint frequency distributions.
- Constructing and interpreting various bar charts.
- Building a stem and leaf diagram.
- Creating a line chart and interpreting trends.
- Constructing and interpreting a scatter diagram.

FQA

1. What is a frequency distribution and how is it used?

A frequency distribution is a summary of data that shows the number of times each distinct category or class appears within a dataset. It helps to understand how the data is spread across different values or groups. For discrete data, you list each possible value and count how many times each value occurs. For continuous data, you need to group values into classes (intervals) and count how many observations fall into each class. This provides a clear view of the data's distribution and the frequency of each value.

2. How do you create a frequency distribution for discrete data, and what is relative frequency?

To create a frequency distribution for discrete data:

1. List all possible values the data can take.
2. Count the occurrences of each value.
3. Calculate relative frequency: Divide the frequency of each value by the total number of observations. This shows the proportion of the total data represented by that value.

For example, if you asked 200 customers how many days a week they read a newspaper, you might find 44 read it 0 days, giving a relative frequency of $44/200 = 0.22$ (22%).

3. What are the main considerations when creating a frequency distribution for grouped (continuous) data?

When creating frequency distributions for continuous data or discrete data with many possible outcomes, you group the data into classes. Key considerations include:

- **Mutually Exclusive Classes:** Classes should not overlap (e.g., 0-9, 10-19, not 0-10, 10-20).
- **All-Inclusive Classes:** The set of classes must contain all data values.
- **Equal Class Widths:** It is preferable if each class has equal width, which is the difference between the highest and lowest values within each class.
- **Avoid Empty Classes:** Avoid classes with no data points falling into them.
- **Determining the Number of Classes:** Use a rule of thumb that the number of classes should be between 5 and 20. The $2k \geq n$ rule (where k is the number of classes and n is the number of data values) can also help determine this.

4. What is a frequency histogram and what kind of information does it provide?

A frequency histogram is a graph that displays a frequency distribution. It has the possible outcomes for a variable on the horizontal axis and the frequency of each outcome on the vertical axis. The key feature of a histogram is that there are no gaps between the bars, to represent that the data is continuous. Histograms help visualize the following:

- **Centre of the Data:** The general location of the data.
- **Spread of the Data:** How the data is dispersed or variable.
- **Shape of the Data:** The overall distribution pattern of the data (e.g., symmetrical, skewed).

5. How do cumulative frequency and cumulative relative frequency differ from standard frequency?

Standard frequency displays the number of times a value or a class appears within the data. Cumulative frequency shows the *total* number of observations with values less than or equal to the *upper limit* of each class. Cumulative relative frequency shows the *proportion* of observations with values less than or equal to the *upper limit* of each class. Cumulative versions track how the frequency or proportion builds up as you move through the dataset.

6. What is a joint frequency distribution and when would you use one?

A joint frequency distribution is used when you have data characterised by *more than one variable*. It is a table that shows the number of observations for each combination of values across the variables, and it can be constructed for both qualitative and quantitative data. For instance, a joint frequency distribution might look at the numbers of customers who paid by charge and also used car park number 1. It's useful when you want to investigate the relationship between different variables.

7. What are the key features of bar charts, pie charts, stem and leaf diagrams, line charts and scatter diagrams and how are they used?

These different chart types display different types of data:

- **Bar Charts:** Use rectangles to represent categorical data, where the height or length of each bar shows the frequency or percentage of a particular category. The bars can be vertical or horizontal.
- **Pie Charts:** Use a circle divided into "slices" where the size of each slice represents the proportion of a total associated with each category.

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- **Stem and Leaf Diagrams:** Used for quantitative data, this diagram separates each data point into a "stem" (usually the leading digits) and a "leaf" (the trailing digit), providing a quick visual overview of the data's distribution.
- **Line Charts:** Used to show time-series data, displaying the values of a variable over time, and useful for highlighting trends.
- **Scatter Diagrams:** Use a two-dimensional graph that shows the relationship between two quantitative variables. They can help visualize whether there is a correlation between the variables and can help to determine a dependent and independent variable.

8. What is a Pareto chart, and when would you use one?

A Pareto chart is a specific type of bar chart used to prioritize categories of a data set by their frequency or impact, using what is known as the Pareto principle, which says that roughly 80% of effects come from 20% of causes. Bars are ordered from the highest to lowest. Pareto charts help to separate the "vital few" categories from the "trivial many". For example, you might use a Pareto chart to identify the main causes of machine downtime, by ranking the downtime minutes associated with each cause from highest to lowest.

Quiz

Instructions: Answer each question in 2-3 sentences.

1. What is a frequency distribution, and what purpose does it serve?
2. Explain the difference between discrete and continuous data, providing an example of each.
3. What are the three key criteria that must be met when building classes for a grouped frequency distribution?
4. Describe the rule of thumb and the mathematical rule for determining the appropriate number
5. of classes for a grouped frequency distribution.
6. What is the difference between a frequency distribution, a cumulative frequency distribution, and a relative frequency distribution?
7. What is a histogram, and what three pieces of information does it provide?

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8. What is a joint frequency distribution, and how does it differ from a simple frequency distribution?
9. Describe how bar charts and pie charts can be used to represent data.
10. What are the key components of a line chart, and what type of data is it best suited for?
11. Explain the purpose of a scatter diagram and how to identify the dependent and independent variables.

Essay Questions

Instructions: Respond to the following questions in an essay format, demonstrating a thorough understanding of the concepts.

1. Discuss the importance of appropriate class selection when creating a grouped frequency distribution. How do the number and width of classes impact the interpretation of the data?

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2. Compare and contrast histograms and bar charts. When is each type of chart most appropriate, and what types of data do they best represent?
3. Explain how to construct a joint frequency distribution. Provide a detailed example and illustrate how you can use that distribution to draw conclusions about the data.
4. Discuss the various ways in which you can use the line chart to visualize data.
5. Using examples from the provided source material, explain the advantages and disadvantages of using different data visualizations such as bar charts, pie charts, line graphs, and scatter diagrams.

Quiz Answer Key

1. A frequency distribution is a summary of a data set that shows the number of observations within distinct categories or classes. It helps to organise and understand the distribution of values in a dataset, revealing patterns and frequencies.
2. Discrete data can only take on countable values, such as the number of late deliveries per month or student ages. Continuous data can take any value within a range, like weight, length, or time.
3. Classes must be mutually exclusive (no overlap), all-inclusive (contain all possible data values), and ideally, of equal width. These criteria ensure that data is correctly classified without ambiguity.
4. The rule of thumb suggests using between 5 and 20 classes. The mathematical rule uses $2k \geq n$, where k is the number of classes, and n is the number of data values. This formula suggests the minimum number of classes, k , based on the size of your data set.
5. A frequency distribution shows the counts for each category or class. A cumulative frequency distribution shows the number of observations less than or equal to the upper limit of each class. Relative frequency distribution shows the proportion of observations in each category compared to the total number of observations.
6. A histogram is a graph that displays a frequency distribution where the horizontal axis represents possible outcomes, and the vertical axis shows frequencies. It reveals the center of the data, its spread, and its shape.
7. A joint frequency distribution shows the frequencies of combinations of two or more variables. Unlike simple frequency distributions that consider only one variable, it shows how multiple variables co-occur.
8. Bar charts use rectangles (bars) to represent categorical data, where height or length indicates the frequency or percentage. Pie charts show the proportion of each category as slices of a circle, illustrating parts of a whole.
9. A line chart displays time on the horizontal axis and the variable of interest on the vertical axis, where points are connected by a straight line. It's suitable for displaying time-series data to show trends and changes over time.
10. A scatter diagram is used to show the relationship between two quantitative variables by plotting data points on a graph. The dependent variable is the one thought to be influenced by the other (independent variable).

Glossary of Key Terms

1. **Bar Chart:** A graphical representation of categorical data using rectangles (bars) where the length or height of each bar represents the frequency, percentage, or another measure associated with the category.
2. **Class Boundaries:** The upper and lower values defining each class in a grouped data frequency distribution.
3. **Cumulative Frequency Distribution:** A summary of data displaying the number of observations with values less than or equal to the upper limit of each class.
4. **Cumulative Relative Frequency Distribution:** A summary of data displaying the proportion of observations with values less than or equal to the upper limit of each class.
5. **Continuous Data:** Data that can take any value within a given range; also known as uncountable data. For example, weight, length, and time.
6. **Dependent Variable:** In a scatter diagram, this is the variable whose values are thought to be a function of, or influenced by, the other variable.
7. **Discrete Data:** Data that can only take on a countable number of possible values; for example, student ages, number of product categories purchased.
8. **Frequency Distribution:** A summary of a data set showing the number of observations within each distinct category or class.
9. **Frequency Histogram:** A graph displaying a frequency distribution, where the horizontal axis contains possible outcomes and the vertical axis shows frequencies, represented by bars with no gaps between them.
10. **Grouped Data Frequency Distribution:** A summary of data where data is organized into classes, usually used with continuous data or discrete data with many possible outcomes.
11. **Independent Variable:** In a scatter diagram, this is the variable whose values are thought to impact the values of the dependent variable.
12. **Joint Frequency Distribution:** A table showing the frequencies of the occurrences of combinations of two or more variables.
13. **Line Chart:** A two-dimensional chart showing time on the horizontal axis and the variable of interest on the vertical axis. It is used to display time-series data.
14. **Pareto Chart:** A specific type of bar chart used to separate the "vital few" from the "trivial many," with bars representing categorical values and organized from high to low.

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15. **Pie Chart:** A circular graph divided into "slices" to represent proportions of a total, where the size of each slice corresponds to the magnitude of the displayed variable associated with each category.
16. **Relative Frequency:** The proportion of total observations within a given category.
17. **Scatter Diagram (Scatter Plot):** A two-dimensional graph of plotted points, where each axis represents a quantitative variable, used to show relationships between two variables.
18. **Stem and Leaf Diagram:** A diagram used to present quantitative data in a way that shows the distribution of values while retaining the original values.

Chapter 3:

Briefing Document

Key Learning Outcomes

The chapter aims to equip the reader with the ability to:

- **Calculate Central Tendency:** Compute the mean, median, mode, and weighted mean and understand their significance.
- **Visualise Data Spread:** Construct and interpret box-and-whisker plots.
- **Quantify Data Dispersion:** Compute the range, interquartile range (IQR), variance, and standard deviation and understand what they represent.
- **Apply Standardisation:** Calculate z-scores and the coefficient of variation (CV) and use them for decision-making.
- **Understand Data Distribution:** Apply the Empirical Rule and Tchebysheff's Theorem.
(Note: The excerpts only discuss the Empirical Rule, not Tchebysheff's Theorem)

Core Concepts

Measures of Center and Location

- **Parameter vs. Statistic:** A **parameter** is a measure computed from an *entire population* and is constant as long as the population doesn't change.
- "A measure computed from the entire population – As long as the population does not change, the value of the parameter will not change."
- A **statistic** is calculated from a *sample* drawn from a population and its value will vary depending on the sample chosen.
- "A measure computed from a sample that has been selected from a population – The value of the statistic will depend on which sample is selected."
- **Measuring Centre:** Aims to describe the typical value in a dataset.
- **Measuring Spread:** Aims to quantify how much the data values vary around the centre.

Mean

- **Definition:** The arithmetic average; sum of all values divided by the number of values.
- **Population Mean (μ):** Formula: $\mu = (\sum x_i) / N$ (sum of individual values divided by the population size).
- **Sample Mean ($\bar{x}$):** Formula: $\bar{x} = (\sum x_i) / n$ (sum of individual values divided by the sample size).
- **Properties:** Uses all the data.
- Sensitive to extreme values (outliers) .
- The 'balance point' of the data: "Sum of Deviations from the mean always equals zero".
- **Example:** The chapter illustrates the calculation of population and sample means using examples like a hospital census and sales data. It demonstrates how changing a single extreme value can drastically impact the mean.

Median

- **Definition:** The midpoint of the data when arranged in numerical order (a 'data array'). "The median is a center value that divides a data array into two halves".
- **Median Index:** Used to find the median's position in a data array: $i = (n+1)/2$
- **Properties:** Not affected by extreme values.
- Uses only the middle value(s).
- 50% of data are above and 50% are below the median.
- **Calculation:** For odd n , it's the middle value; for even n , it's the average of the two middle values.
- **Examples:** Examples use both sales figures and salary data to show that the median is stable even when data contain extreme values.

Mode

- **Definition:** The most frequently occurring value in a data set. "The value in the data which appears most frequently"
- **Properties:** Not affected by extreme values.
- Can be used with both quantitative and qualitative data.
- Can have more than one mode (bimodal) or no mode.

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- **Example:** Uses sales data and NBA player weights to show mode calculation, and notes that it can be a weak measure of the center of a dataset.

Weighted Mean

- **Definition:** The mean of values that have been assigned weights based on their relative importance.
- **Application:** The chapter uses financial portfolio data to illustrate its use.

Percentiles and Quartiles

- **Percentiles:** A value below which a given percentage of data falls. "p% are less than or equal to this value. (100 – p)% are greater than or equal to this value."
- **Quartiles:** Divide the dataset into four equal parts:
 - Q1 (1st quartile): 25th percentile
 - Q2 (2nd quartile): 50th percentile (also the median)
 - Q3 (3rd quartile): 75th percentile
- **Calculation:** The chapter outlines the step-by-step procedure to calculate percentiles, involving ordering the data and calculating a location index.

Box and Whisker Plot

- **Definition:** A visual representation of data distribution using quartiles, the median, and outliers.
- "The box has a width that ranges from the first quartile (Q1) to the third quartile (Q3)."
- **Construction:** The chapter outlines the steps to construct a boxplot: Calculate Q1, Q2, Q3, identify outliers, and determine upper and lower limits.
- $Lower\ Limit = Q1 - 1.5 (Q3 - Q1)$
- $Upper\ Limit = Q3 + 1.5 (Q3 - Q1)$
- **Application:** The example of miles between fill-ups for a vehicle highlights the use of boxplots.

Data Level Issues

- **Importance:** The level of data is vital when choosing which measures to calculate.

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- **Mean:** Appropriate for ratio or interval level data (e.g., age, income), but not for ordinal or nominal data (e.g., education level or marital status). "Computing a Mean for Ordinal Data – Not Recommended".
- **Median:** Suitable for ratio and interval data, and is acceptable for ordinal data. Not for nominal data.
- **Mode:** Can be used for all data types.

Measures of Dispersion/Variation

- **Purpose:** To quantify how spread out or variable the data is around the centre.

Range

- **Definition:** The difference between the maximum and minimum values.
- $R = \text{Maximum Value} - \text{Minimum Value}$
- **Properties:** Simple to compute.
- Uses only two values.
- Very sensitive to extremes.
- **Example:** The chapter uses staff years of experience to demonstrate the calculation of range.

Interquartile Range (IQR)

- **Definition:** The difference between the third and first quartiles.
- $\text{Interquartile Range} = Q3 - Q1$
- **Properties:** Eliminates problems caused by outliers.
- Ignores some extreme values.

Variance

- **Definition:** The average of the squared deviations from the mean.
- **Population Variance (σ^2):** Formula: $\sigma^2 = \sum (x_i - \mu)^2 / N$
- **Sample Variance (s^2):** Formula: $s^2 = \sum (x_i - \bar{x})^2 / (n-1)$
- **Note:** Variance is expressed in squared units.

- **Example:** Used production volumes from different departments to show the importance of measures of spread.

Standard Deviation

- **Definition:** The square root of the variance.
- **Population Standard Deviation (σ):** Formula: $\sigma = \sqrt{\sigma^2}$
- **Sample Standard Deviation (s):** Formula: $s = \sqrt{s^2}$
- **Properties:** The most commonly used measure of variation.
- Has the same units as the original data.
- Indicates an 'average distance' of data points from the mean.
- **Example:** Uses convention center attendance and restaurant sales to show standard deviation.

Using Mean and Standard Deviation Together

Coefficient of Variation (CV)

- **Definition:** The ratio of the standard deviation to the mean, expressed as a percentage.
"The coefficient of variation is used to measure variation relative to the mean."
- Population CV = $(\sigma / \mu) * 100\%$
- Sample CV = $(s / \bar{x}) * 100\%$
- **Properties:** Measures relative variation, allows comparisons of datasets with different units or different scales.
- **Example:** The chapter uses two stocks with the same standard deviation but different average prices to show why CV can be important.

Empirical Rule

- **Description:** For a bell-shaped (normal) distribution, approximately:
- 68% of the data falls within one standard deviation of the mean ($\mu \pm 1\sigma$).
- 95% of the data falls within two standard deviations of the mean ($\mu \pm 2\sigma$).
- Almost all data falls within three standard deviations of the mean ($\mu \pm 3\sigma$).
- **Example:** Used Uber travel distances to illustrate the application of the Empirical Rule.

Standardised Data Values (Z-scores)

- **Definition:** The number of standard deviations a value is from the mean.
- Population z-score = $(x - \mu) / \sigma$
- Sample z-score = $(x - \bar{x}) / s$
- **Application:** Used to compare values from different datasets with different means and standard deviations.
- **Example:** Used IQ scores and University Entrance Tests to show how Z-scores allow you to determine relative standing.

Conclusion

Key takeaway points include:

- There are various measures for the 'center' and 'spread' of data.
- The mean is sensitive to extreme values, whereas the median is not.
- The standard deviation and variance measure data dispersion from the mean.
- Standardised data (z-scores) allow for comparison across datasets and can determine position relative to the mean.
- The Coefficient of Variation allows the comparison of variation across different datasets with different units.

FQA

1. What are the key measures of central tendency and how do they differ?

The key measures of central tendency are the mean, median, and mode. The **mean** is the arithmetic average of all values, calculated by summing all the data points and dividing by the total number of data points. It is sensitive to extreme values and is the balance point of the data. The **median** is the middle value in an ordered data set; half the data is above and half below. It is not affected by extreme values, making it useful when data may be skewed. The **mode** is the value that appears most frequently in the data. It can be used for both numerical and categorical data, but it may not accurately reflect the centre if multiple modes exist or when the most frequent value is an outlier.

2. What is the difference between a parameter and a statistic?

A **parameter** is a measure calculated from an entire population. As long as the population does not change, its value remains constant. A **statistic**, on the other hand, is a measure calculated from a sample taken from a population. The value of a statistic will depend on which sample is selected, making it a variable estimate of a population parameter.

3. What are quartiles and percentiles, and how are they calculated?

Percentiles divide a data set into 100 equal parts. The p th percentile is the value below which $p\%$ of the data falls. **Quartiles** are specific percentiles that divide a dataset into four equal parts: the first quartile (Q1) is the 25th percentile, the second quartile (Q2) is the 50th percentile (also the median), and the third quartile (Q3) is the 75th percentile. To calculate them, data must first be sorted in ascending order. A percentile location index is calculated based on the desired percentile. If the index is not an integer it's rounded up to the next highest integer. If the index is an integer, the percentile value is the average of values at index location i and $i + 1$.

4. What is a box and whisker plot, and how is it constructed and interpreted?

A box and whisker plot, also called a boxplot, is a graphical way to visualise the distribution of data, mainly through the quartiles. The box itself extends from the first quartile (Q1) to the third quartile (Q3), with a line representing the median (Q2) within the box. "Whiskers" extend from the box to the smallest and largest data points within 1.5 times the interquartile range (IQR). The IQR is the difference between Q3 and Q1 ($Q3 - Q1$). Data points falling outside of the "whiskers" are considered outliers, which are typically marked with an asterisk. The plot helps to visualise the spread, skewness and identify outliers of a data distribution.

5. What are the key measures of dispersion (or variation) and how are they different from measures of central tendency?

Key measures of dispersion or variation show how spread out the data points are around the central tendency. The main dispersion measures are the range, interquartile range (IQR), variance, standard deviation and coefficient of variation (CV). The **range** is the difference between the maximum and minimum values. The **IQR** is the range of the middle 50% of the data. The **variance** measures the average squared distance of data points from the mean. The **standard deviation** is the square root of the variance and is the most commonly used measure of variation, expressed in the same units as the data. The **coefficient of variation** (CV) is the standard deviation divided by the mean, which allows the comparison of variability

between datasets with differing units or magnitudes. Measures of central tendency, like the mean, median and mode, describe the typical, average or center of the dataset, whilst measures of dispersion quantify how far from that center individual values can be.

6. What is the Empirical Rule, and when can it be applied?

The Empirical Rule, sometimes called the 68-95-99.7 rule, is applicable for data sets with bell-shaped distributions. It states that approximately 68% of the data values fall within one standard deviation of the mean, 95% of the values within two standard deviations, and virtually all (99.7%) of the values within three standard deviations of the mean. It helps to quickly understand data spread and distribution in specific contexts.

7. What is a z-score (standardized data value) and how is it calculated?

A z-score (or standardized data value) measures how many standard deviations a particular data point is away from the mean of its distribution. It's calculated by subtracting the mean from the data value and then dividing by the standard deviation. For a population, $z = (x - \mu) / \sigma$, where x is the data value, μ is the population mean, and σ is the population standard deviation. For a sample, $z = (\bar{x} - \bar{x}) / s$, where x is the sample data value, $\bar{x}$ is the sample mean, and s is the sample standard deviation. The z-score allows the comparison of data points from different datasets, and helps identify whether a specific data point is considered typical, or extreme.

8. How does the level of data affect which descriptive statistics should be used?

The level of data is critical for determining which descriptive statistics can be meaningfully calculated and interpreted. For **ratio** or **interval** data (data with equal intervals and with or without a true zero point), all measures are appropriate, including the mean, median, mode, standard deviation and variance. For **ordinal** data (data with a meaningful order but not equal intervals), the median and mode can be used, but the mean might be misleading. For **nominal** data (categorical data with no inherent order), only the mode is appropriate. Calculations such as the mean should be avoided for ordinal or nominal data because it can lead to meaningless results.

Quiz

Answer the following questions in 2-3 sentences each.

1. What is the difference between a parameter and a statistic?
2. Explain what is meant by the "median" of a dataset.
3. What is the main advantage of using the median instead of the mean as a measure of central tendency?
4. Give an example of a situation where the mode would be a particularly useful measure of central tendency.
5. What does the interquartile range (IQR) measure, and why is it useful?
6. Why do we square the deviations from the mean when calculating variance?
7. Explain the purpose of the coefficient of variation (CV).
8. State the main idea of the Empirical Rule, and under what condition it applies.

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9. What is a z-score, and how does it relate to standard deviations?
10. Why should you avoid calculating the mean of nominal data?

Essay Questions

Answer each of the following questions in essay format, using concepts and terms from the source material.

1. Discuss the strengths and weaknesses of the mean, median, and mode as measures of central tendency. When is each measure most appropriate to use, and why?
2. Explain how the range, interquartile range, variance, and standard deviation each provide different insights into the dispersion of a dataset. Describe the advantages of using each of these measures.
3. Describe the construction of a box-and-whisker plot. How can this graphical tool be used to identify and interpret central tendency, dispersion, and outliers within a dataset?
4. Explain how the Empirical Rule is used and applied to interpret data. How can knowledge of data distribution shapes (symmetric or skewed) affect the interpretations?

5. Describe the purpose of the z-score. How might standardized data values and the coefficient of variation aid in making more informed decisions? Give specific examples to support your answers.

Quiz Answer Key

1. A parameter is a measure calculated from the entire population, whereas a statistic is calculated from a sample of the population. The value of a parameter does not change unless the population changes, while a statistic's value varies depending on the sample selected.
2. The median is the midpoint of a dataset when it is ordered from lowest to highest value. It divides the data into two equal halves, with 50% of the data points falling above it and 50% below.
3. The median is not affected by extreme values, which makes it a more robust measure of central tendency than the mean when dealing with skewed data or datasets with outliers.
4. The mode is useful when looking for the most frequently occurring category of data. For example, when determining the most common shoe size sold in a shop.
5. The interquartile range (IQR) measures the spread of the middle 50% of the data. It is useful because it is not affected by extreme values and provides a more robust measure of dispersion than the simple range.
6. Squaring the deviations from the mean ensures that all deviations are positive and gives more weight to larger deviations from the mean when calculating the variance. This helps capture the spread of the data around the mean.
7. The coefficient of variation (CV) is a measure of relative variability. It expresses the standard deviation as a percentage of the mean, allowing for the comparison of variability across datasets with different means and/or different units of measurement.
8. The Empirical Rule states that for bell-shaped distributions, approximately 68% of the data falls within one standard deviation of the mean, 95% within two standard deviations, and virtually all data within three standard deviations of the mean.
9. A z-score indicates the number of standard deviations a specific data point is from the mean. It standardises data, allowing comparisons across different distributions, with positive z-scores representing values above the mean and negative z-scores values below the mean.

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10. Calculating the mean for nominal data, where values represent categories without intrinsic numerical order, is inappropriate as it implies an order or quantity which is non-existent. Means on this type of data are meaningless and misleading.

Glossary of Key Terms

1. **Parameter:** A measure computed using data from the entire population.
2. **Statistic:** A measure computed using data from a sample taken from a population.
3. **Mean:** The arithmetic average of a set of data, calculated by summing all values and dividing by the number of values. Also known as the average.
4. **Median:** The middle value in an ordered dataset, dividing the data into two equal halves.
5. **Mode:** The value that appears most frequently in a dataset.
6. **Weighted Mean:** The mean of a dataset where each value is given a certain weight, reflecting its relative importance.
7. **Data Array:** A list of data that has been arranged in numerical order.
8. **Interquartile Range (IQR):** The difference between the third quartile (Q3) and the first quartile (Q1) of a dataset; measures the spread of the middle 50% of the data.
9. **Range:** The difference between the maximum and minimum values in a dataset; a simple measure of dispersion.
10. **Variance:** The average of the squared differences between each data value and the mean; a measure of data dispersion in units squared.
11. **Standard Deviation:** The square root of the variance; a measure of the typical deviation from the mean, in the same units as the data.
12. **Coefficient of Variation (CV):** A measure of relative variability, calculated by dividing the standard deviation by the mean, expressed as a percentage; allows for comparison of dispersion across datasets with different means and/or units.
13. **Empirical Rule:** A statistical rule stating that, for bell-shaped distributions, approximately 68% of the data values fall within one standard deviation of the mean, 95% within two standard deviations, and virtually all values within three standard deviations.
14. **Z-score:** The number of standard deviations a data value is away from the mean. Also known as a standardised data value. It quantifies how unusual a data point is in the context of its distribution.
15. **Percentiles:** Values that divide a dataset into 100 equal parts. The p th percentile is a value below which p percent of the data falls.

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16. **Quartiles:** Values that divide a dataset into four equal parts. The first quartile (Q1) is the 25th percentile, the second quartile (Q2) is the 50th percentile (median), and the third quartile (Q3) is the 75th percentile.
17. **Box-and-Whisker Plot:** A graphical representation of a dataset using a box to represent the IQR, a line for the median, and whiskers that extend to the data's limits, with outliers marked with asterisks.
18. **Outlier:** A data point that falls significantly outside the expected range of values in a dataset.
19. **Nominal Data:** Data that represents categories with no intrinsic numerical order or ranking.
20. **Ordinal Data:** Data that represents categories with a meaningful rank order, but with non-uniform differences between the ranks.
21. **Interval Data:** Data that is measured on a scale with uniform intervals, but has an arbitrary zero point.
22. **Ratio Data:** Data that is measured on a scale with uniform intervals, and has a meaningful zero point.